Online Appendix to

"AI Exposure or Occupational Composition? Young-Worker Employment Comparisons in the CPS"

Lily Luo

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Contents

A Data Reconstruction and Treatment Contract

A.1 Source files and sample flow

The stock analysis uses two restricted IPUMS CPS extracts. The wide file supplies January 2017–July 2026 records; a second file supplies the five March Basic Monthly samples that replace incorrectly selected March ASEC samples in 2017–2021. The replication manifest records extract identifiers, creation metadata where available, byte sizes, and SHA-256 hashes without redistributing microdata. The March replacement occurs before eligibility filters or occupation routing. The superseded rows contribute no positive-final-weight analysis records; the replacement adds 252,862 eligible records.

The wide file is IPUMS extract 9: its request was submitted on August 25, 2026, and the completed-file receipt was recorded four minutes later. Separate hashes authenticate its submitted specification, data, DDI, and basic codebook. The March repair is extract 11; its actual request, returned API response, data, DDI, and basic-codebook hashes are verified, but its surviving API response contains no submission or completion timestamp, so none is inferred from filesystem metadata. Flow models additionally use extract 10 for `LNKFW1MWT`; it was recorded complete when checked on August 31, 2026, but the check time is not called an API completion time. Exact timestamps and hashes appear in the input manifest and provenance receipt. Restricted files, account metadata, download links, and direct identifiers are not redistributed.

TABLE 1. Calendar and cell reconstruction audit

Object	Count	Audit rule
Expected calendar months	115	January 2017–July 2026
Observed months	114	October 2025 not collected
Static months	113	December 2022 transition excluded
Primary occupations	468	named fixed support
Complete occupation–month grid	52,884	468×113
Positive-total fitted cells	51,891	includes one-sided zeros
Both-age-zero cells	993	no likelihood term
One-sided zero cells retained	12,965	finite boundary contributions

Notes: October 2025 is absent, not coded zero or interpolated. December 2022 is observed but excluded from the static pre/post fit.

Eligibility requires age 18–65, employment, a valid occupation, and positive `WTFINL`. The analysis scans 9,843,021 source rows, retains 5,322,047 eligible source records, and creates

6,188,956 route-expanded descendants. The last count is fractional before 2020 and is not a count of distinct respondents. The principal age groups are formed after routing. Of the complete grid, 13,883 young cells and 1,068 older cells have zero respondent-equivalent records. These sparse-cell facts motivate the finite-sample audit; they do not justify selecting cells on realized young counts.

A.2 Reconstructed preperiod treatment

The substantive treatment is rebuilt using only January 2017–November 2022 stock. On the fixed 468-occupation support, the employment-weighted beta mean is 0.37520 and standard deviation is 0.21041. The tie-preserving cut values are 0.153846, 0.324615, 0.456522, and 0.537037. Webb software exposure is standardized over the same 71-month construction window. Table 2 compares the rebuilt contract with the historical production contract.

TABLE 2. Historical and reconstructed treatment contracts

Contract	Construction window	Post stock used	Q5–Q1 coefficient
Historical production	full static panel	Yes	−0.134554
Reconstructed	Jan. 2017–Nov. 2022	No	−0.132109

Notes: Both coefficients use the corrected 113-month outcome calendar and identical occupation support. Nine occupations change quintile. The reconstructed-minus-historical paired difference is 0.002445 with 95-percent interval $[-0.001269, 0.006158]$.

The reconstructed contract contains 129, 99, 82, 67, and 91 occupations in Q1–Q5. Corresponding preperiod stock shares are 20.06, 19.96, 20.16, 20.31, and 19.51 percent. The complete membership file identifies all nine occupations whose historical and rebuilt labels differ. All revised substantive models use the reconstructed labels unless explicitly described as a historical audit.

A.3 Survey-production dates

Official BLS documentation establishes population-control discontinuities in January 2025 and January 2026. BLS did not revise the preceding official months onto the new bases and does not publish counterfactual age-by-occupation weights. October 2025 was not collected; November collection began late, used reduced overlap and a modified weighting procedure, and had a lower response rate. IPUMS later processed the reissued January 2026 file and corrections to longitudinal identifiers. The stock and flow manifests preserve their respective extract dates and hashes. Endpoint and exclusion checks diagnose reliance on these dates, but no aggregate adjustment factor is imposed on detailed cells.

B Occupational Mapping and Analysis Support

B.1 Bridge construction

The early panel reports Census-2010 occupations and the later panel Census-2018 occupations. Exposure inputs additionally span SOC vintages. The versioned bridge first routes 503 early source codes into 568 target codes. Fifty-six sources have more than one target, with maximum multiplicity seven. Official shares partition each source record's final weight; separate young and older routing would require validation data that the public adjacent-vintage files do not provide. The routed early and current stocks reconcile to their sources with relative error below 2×10^{-16}.

In the frozen mapping-only lookup, the equal-administrative AIOE variant has full-route coverage for 480 of the 503 pre-2020 source codes. This 480-code mapping count is not the regression sample: intersecting the rebuilt beta and Webb inputs with positive young preperiod stock yields the 468-occupation primary analysis support below.

The primary joint beta/Webb universe contains 468 occupations and 87.19 percent of stock in the audited candidate universe. Table 3 gives mutually exclusive exclusion counts. A machine-readable file lists each included and excluded occupation, its preperiod stocks, scores, and reason. This file, rather than a hard-coded count in the paper, defines support.

TABLE 3. Primary-support exclusions

Reason	Occupations
Missing Rule-A beta only	41
Missing Webb software only	22
Missing both exposure inputs	8
Nonpositive young preperiod stock	2
Total excluded	73
Primary retained	468

B.2 Separating score correction from population expansion

A literal exact-code merge between SOC-2010 AIOE and post-2018 employment data leaves only 3.33 percent of computer-and-mathematical employment. The version-aware route reaches 97.7 percent. The four-row continuous comparison separates the two operations that an aggregate "crosswalk effect" would conflate:

TABLE 4. AIOE taxonomy repair and support expansion

Score	Support	Occupations	Continuous coefficient
Literal original	original	410	−0.01885
Repaired	same original	410	−0.01920
Repaired	expanded	495	−0.03156
Repaired	expanded, excluding SOC15	—	−0.02940

Notes: The first row is an invalid vintage merge retained as an implementation audit, not a defensible specification. Row 1 to 2 isolates score repair on fixed support; row 2 to 3 changes the population; the last row tests whether readmitted computer-and-mathematical occupations alone account for that expansion movement. The continuous-score scale is held fixed within the audit.

The invalid merge is documented as a failure encountered in this project. I do not claim that a published paper made the same error without direct verification of its code and occupational universe.

B.3 Split-route and age-allocation sensitivity

Within the primary support, split sources contribute 14.93 percent of young and 17.75 percent of older early stock. Ninety-nine target occupations can receive a split route. A one-to-one-target restriction and an aggregation to stable Census-2010 categories both retain negative historical-contract estimates, but each changes the target population. They are mapping sensitivities, not repairs of the primary bridge.

Using the same route proportions by age is transparent but unvalidated. Accounting ranges that allocate every ambiguous source in all feasible ways place the 2017–2019 young-to-older stock ratio between 0.1031 and 0.1480 in Q1 and between 0.0790 and 0.1175 in Q5. These are stock-accounting bounds, not bounds on the nonlinear regression coefficient. A limited odds-tilt sensitivity changes the historical-contract coefficient only modestly because 6.88 percent of early stock is eligible, but it cannot validate age-specific routing. No adjacent coding vintage is treated as a same-person validation sample.

C Broad-Family Support, Profiles, and Influence

C.1 Family-by-quintile support

Table 5 reports, for every SOC two-digit family, the number of retained occupations, preperiod stock share, exposure range, quintiles present, direct-tail overlap, and nuisance-adjusted Q5 information share. Only arts/media (SOC27), health practitioners (SOC29), health support (SOC31), and sales (SOC41) span Q1 and Q5 under the reconstructed comparison. Other families connect the Q5–Q1 regression coefficient through intermediate quintiles and the imposed common Q2–Q5 slopes.

TABLE 5. Broad-family exposure support

SOC2	Family	Occs.	Stock (%)	Beta range	Quintiles	Q1/Q5	Q5 info. (%)
11	Management	29	13.56	0.367–0.586	3|4|5	No	3.04
13	Business and Financial Operations	28	6.22	0.405–0.636	3|4|5	No	5.99
15	Computer and Mathematical	15	4.31	0.500–0.950	4|5	No	0.28
17	Architecture and Engineering	19	2.28	0.406–0.727	3|4|5	No	1.87
19	Life, Physical, and Social Science	21	1.25	0.344–0.723	3|4|5	No	0.54
21	Community and Social Service	12	1.31	0.111–0.439	1|2|3	No	0.20
23	Legal	4	1.34	0.425–0.571	3|4|5	No	0.28
25	Educational Instruction and Library	7	4.53	0.229–0.567	2|3|5	No	3.20
27	Arts, Design, Entertainment, Sports, and Media	25	1.80	0.000–0.958	1|2|3|4|5	Yes	4.86
29	Healthcare Practitioners and Technical	37	6.38	0.067–0.618	1|2|3|4|5	Yes	3.61
31	Healthcare Support	14	3.66	0.074–0.875	1|2|3|5	Yes	7.26
33	Protective Service	17	2.28	0.036–0.415	1|2|3	No	1.42
35	Food Preparation and Serving Related	10	3.27	0.000–0.511	1|2|4	No	10.84
37	Building and Grounds Cleaning and Maintenance	6	2.51	0.000–0.355	1|2|3	No	1.29

Continued on next page

Table 5 continued

SOC2	Family	Occs.	Stock (%)	Beta range	Quintiles	Q1/Q5	Q5 info. (%)
39	Personal Care and Service	17	2.51	0.015–0.400	1|2|3	No	2.97
41	Sales and Related	17	10.06	0.133–0.806	1|2|3|4|5	Yes	17.40
43	Office and Administrative Support	47	11.21	0.216–0.975	2|3|4|5	No	18.92
45	Farming, Fishing, and Forestry	5	0.15	0.043–0.292	1|2	No	0.14
47	Construction and Extraction	30	6.14	0.000–0.417	1|2|3	No	1.97
49	Installation, Maintenance, and Repair	31	3.56	0.000–0.410	1|2|3	No	2.64
51	Production	52	4.14	0.000–0.438	1|2|3	No	5.67
53	Transportation and Material Moving	25	7.52	0.031–0.413	1|2|3	No	5.59

Notes: Stock is each family's share of preperiod employment on the 468-occupation rebuilt support. Q5 information is the family's share of nuisance-adjusted fitted information for the family-conditioned Q5 target; it is neither a sampling-cluster share nor pre-outcome residual-exposure support. Beta ranges and memberships use the corrected preperiod treatment contract. Only Q1/Q5 rows enter the changed-population direct-tail exercise.

Table 6 makes the loss and concentration of identifying information explicit. Family-by-month conditioning retains 29.7 percent of the pooled Q5 target's nuisance-adjusted information, while the direct-tail comparison has only 6.2 effective occupations and its five largest occupation contributions supply 77.8 percent of information. Its normal-theory MDE_{80} is 0.4576 log point. The continuous within-family companion is more diffuse and has $\mathrm{MDE}_{80} = 0.0313$ per one within-family standard deviation, but it imposes a common slope across families.

The direct-tail population retains only occupations in the four spanning families that belong to Q1 or Q5. It contains 29 occupations and 5.03 percent of primary preperiod stock. In the SOC2-by-calendar-month model its coefficient is 0.1494, with interval $[-0.1694, 0.4682]$. This benchmark is valuable because it exposes direct overlap; it is not a preferred estimate for the original population, and the interval includes large effects of either sign.

C.2 Full exposure profile

The pooled Q2–Q5 coefficients are jointly different from zero, but pairwise movements do not establish a monotone dose response. After SOC2-by-calendar-month conditioning, the four

TABLE 6. Family-support information and precision diagnostics

Target and structure	Occ.	I (millions)	I/I_{raw}	I/I_{parent}	Effective occ.	Top-five share	Normal MDE_{80}
Q5–Q1, pooled	468	26.289	0.378	1.000	43.2	0.245	0.1266
Q5–Q1, SOC2 × month	468	7.803	0.112	0.297	53.3	0.218	0.1998
Direct-tail Q5–Q1, SOC2 × month	29	0.669	0.080	0.308	6.2	0.778	0.4576
Continuous within-SOC2 slope, SOC2 × month	468	261.278	0.583	0.965	48.0	0.249	0.0313

Notes: I is nuisance-adjusted Fisher information after projecting the target score on the model's nuisance score; I_{raw} is the raw weighted treatment sum of squares. "Parent" is the corresponding pooled model on the same target and population, so the direct-tail and continuous rows each use their own pooled parent. Because target scales and populations differ, absolute I is not comparable across all rows. Effective occupations are the inverse Herfindahl count of occupation-level information contributions; the top-five share is the sum of the five largest contributions. Fitted information and its effective count are design-concentration diagnostics: neither replaces the nominal occupation-cluster count nor measures first-stage CPS sampling precision. $\text{MDE}_{80} = 2.80\widehat{SE}$ uses the occupation-cluster standard error and is a normal-theory precision diagnostic, not a rejection result. The direct-tail MDE is 0.4576 log point; the continuous MDE is 0.0313 per one within-family standard deviation (0.1084 raw beta units).

coefficients are jointly indistinguishable from zero. Table 7 reports pointwise and simultaneous intervals and the common-draw movement for every group.

TABLE 7. Exposure profile before and after broad-family monthly paths

	Pooled			SOC2 × month			Paired movement
Group	Coef.	Pointwise 95% CI	Simultaneous 95% CI	Coef.	Pointwise 95% CI	Simultaneous 95% CI	Coef. [95% CI]
Q2	-0.0784	[-0.1546,-0.0023]	[-0.1716,0.0147]	-0.0247	[-0.1228,0.0734]	[-0.1431,0.0937]	0.0537 [-0.0051,0.1126]
Q3	-0.0500	[-0.1224,0.0225]	[-0.1383,0.0383]	0.0169	[-0.0859,0.1197]	[-0.1043,0.1381]	0.0669 [-0.0047,0.1386]
Q4	-0.0896	[-0.1578,-0.0214]	[-0.1737,-0.0055]	-0.0126	[-0.1302,0.1051]	[-0.1510,0.1259]	0.0770 [-0.0138,0.1679]
Q5	-0.1321	[-0.2206,-0.0437]	[-0.2393,-0.0249]	-0.0217	[-0.1607,0.1173]	[-0.1836,0.1403]	0.1104 [0.0107,0.2102]

Notes: Q1 is omitted. Simultaneous intervals use each model's common four-target maximum-$|t|$ critical value. To prevent the same Q5 target carrying seed-dependent pointwise intervals, the Q5 pointwise and paired cells reproduce the declared single-target canonical intervals; its simultaneous cells remain the four-target family-profile result. All rows use rebuilt labels and 9,999 common occupation draws within each declared collection.

The continuous within-family companion subtracts each family's preperiod employment-weighted beta mean and imposes a common linear slope. Under SOC2-by-calendar-month conditioning, the rebuilt coefficient is -0.0025 per 0.1084 raw beta units, with interval $[-0.0240, 0.0191]$ in that standardized within-family unit. The unconditioned continuous coefficient is essentially zero as well. This is a different functional form from the tail contrast.

C.3 Leave-one-family and occupation influence

Leave-one-family-out estimates are computed from the same family-conditioning program and reported with the omitted family's stock and target-information contribution. These rows answer whether a broad comparison is concentrated; they are not used to choose which families belong in the main estimate. Table 8 displays the eight largest absolute movements. Omitting office/administrative support or food preparation moves the coefficient in opposite directions by about 0.04 log point, and every paired deletion interval includes zero.

Figure 1 shows the four direct-tail families, selected by descending nuisance-adjusted Q5 information rather than coefficient sign. Within each exposure tail, the young and older employment stocks are shown separately and indexed to their own 2019 means. The figure therefore exposes which age series drives a tail's relative path without treating either age group as untreated.

Occupation leave-one-out calculations are interpreted in the same way. Fast-food and counter workers have high signed influence because low-exposure service employment recovered differently after the pandemic. Other occupations push in the opposite direction. Historical-contract diagnostics show that deleting the five largest absolute prior influences moves the coefficient toward zero, whereas deleting ten or twenty can make it more negative. Symmetric

TABLE 8. Largest leave-one-family-out movements for the conditioned Q5 target

Omitted SOC2	Family	Preperiod stock (%)	Re-estimated coefficient	Movement [95% paired CI]
43	Office and Administrative Support	11.21	-0.0614	-0.0397 [-0.1131,0.0337]
35	Food Preparation and Serving Related	3.27	0.0174	0.0391 [-0.0292,0.1074]
31	Healthcare Support	3.66	-0.0534	-0.0317 [-0.0772,0.0139]
13	Business and Financial Operations	6.22	0.0099	0.0316 [-0.0006,0.0638]
39	Personal Care and Service	2.51	-0.0457	-0.0240 [-0.0496,0.0015]
25	Educational Instruction and Library	4.53	-0.0351	-0.0134 [-0.0383,0.0115]
17	Architecture and Engineering	2.28	-0.0339	-0.0123 [-0.0330,0.0085]
33	Protective Service	2.28	-0.0096	0.0121 [-0.0050,0.0291]

Notes: The parent SOC2-by-calendar-month coefficient is -0.0217. Families are ranked by the absolute coefficient movement, so this is an outcome-informed influence diagnostic rather than a preferred deletion rule. Quintiles and the continuous centering scale remain fixed; all 22 deletions are available in the machine-readable file.

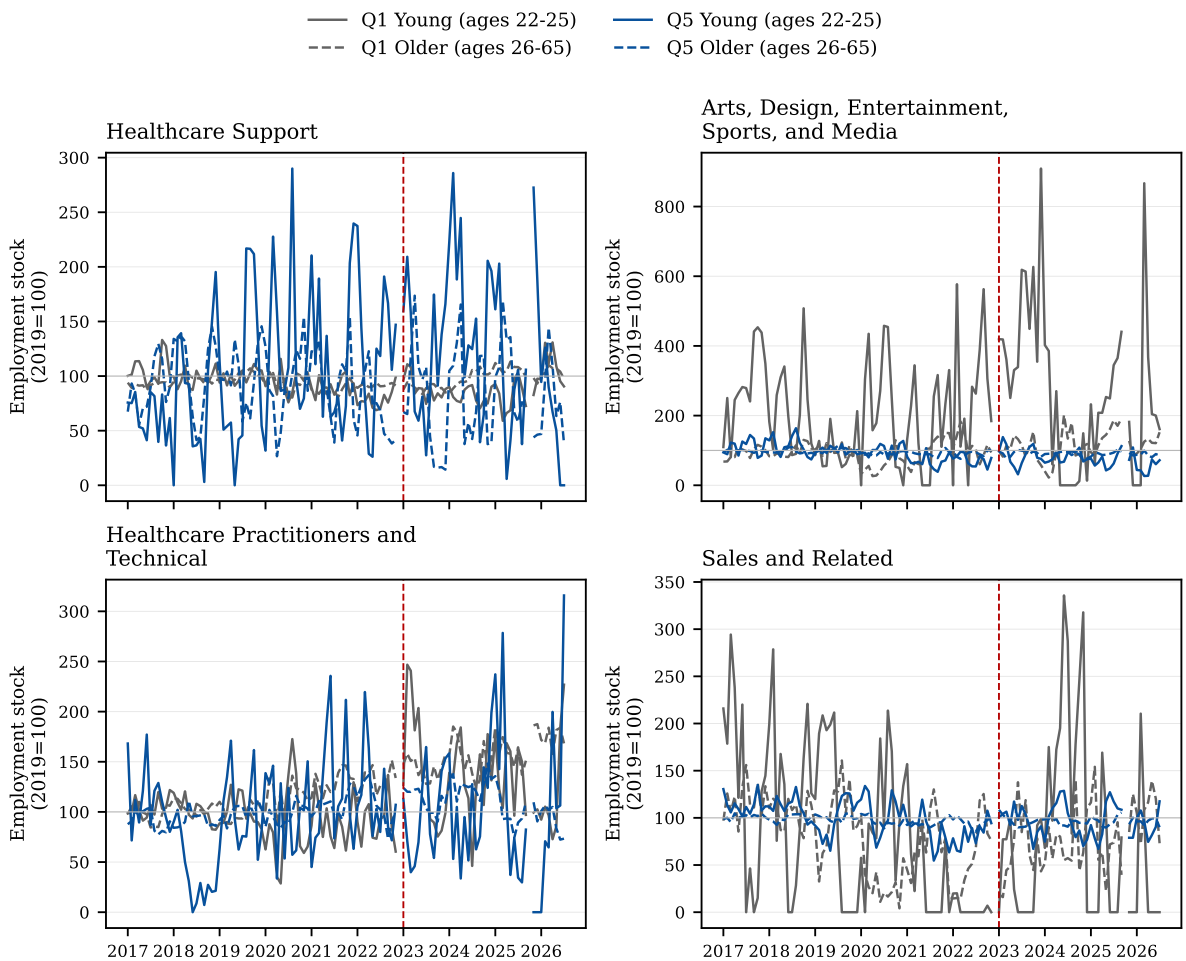

FIGURE 1. Young and older employment stocks in direct-tail families

Notes: Families are selected by direct-tail, SOC2-by-calendar-month Q5 nuisance-adjusted information, with ties broken by SOC2 code; selection does not use coefficient sign. Gray and blue distinguish Q1 and Q5; solid and dashed series distinguish ages 22–25 and 26–65. Each series equals 100 at its own 2019 mean, and the red dashed line marks January 2023. These are descriptive weighted-stock paths without design-based bands.

influence trimming stays near the pooled estimate. Because every such deletion is outcome adaptive, none replaces the broad-family regression.

Economically defined service exclusions are cleaner diagnostics but still change the population. Removing Q1 food-service occupations attenuates the historical-contract absolute coefficient by less than one tenth; removing a broader set of food, building-and-grounds, personal-care, and protective-service occupations does not attenuate it. These results reject a one-occupation explanation, not hypotheses about reopening, immigration, wages, or labor supply that the CPS design does not separately identify.

D Occupational Characteristics and Heterogeneity

D.1 Inputs, timing, and common-support comparisons

The characteristic exercise uses one variable at a time before considering a parsimonious joint model. Every row first re-estimates the unaugmented exposure model on exactly the support available for that characteristic. It then adds the characteristic interacted with young status and the post indicator, reuses the same occupation multipliers, and forms the augmented-minus-baseline difference within each draw. Sample loss and coefficient movement are therefore distinct objects. The corresponding paired minimum detectable effect is the two-sided, five-percent normal-theory quantity $(1.96 + 0.84)SE(\Delta)$; it describes resolution and is never used as a second rejection rule.

The leading computer-use variable is O*NET item 4.A.3.b.1, "Interacting With Computers," from release 24.3 (May 2020). Remotability is the Dingel–Neiman occupation classification. Wage and observed education composition are calculated from the CPS preperiod, while externally coded education requirements are kept conceptually separate. Routine and manual indices use preperiod O*NET inputs. All scales, source files, checksums, weighted correlations, and support counts are stored in the characteristic result directory.

The focused rebuilt-treatment comparison uses the 408 occupations with both computer use and remotability, so every row has identical support. The baseline coefficient is -0.1070 and the computer-use-conditioned coefficient is -0.2124. Their paired difference is -0.1054, with 95-percent interval $[-0.1835, -0.0273]$ and paired $\text{MDE}_{80} = 0.1131$. Thus computer use is a suppressor in this particular regression rather than an attenuating control. This result neither validates the residual as AI nor makes computer use exogenous.

Adding remotability on the same support gives -0.1155; its paired difference from -0.1070 is -0.0085, with interval $[-0.0437, 0.0267]$ and paired $\text{MDE}_{80} = 0.0510$. Adding both computer use and remotability gives -0.2120 and a paired movement of -0.1049, with interval $[-0.1814, -0.0284]$. The other one-at-a-time characteristics likewise do not produce individually detected movements. The expanded module's joint characteristic support falls to 341 occupations and 76.77 percent of rebuilt treatment-construction stock; its wider interval precludes a claim that the combined residual is a purified AI contrast.

D.2 Expanded support-specific and cumulative audit

The separate characteristic module in Tables 9 and 10 reports the support-specific comparisons and cumulative sequence. Its one-at-a-time rows use each characteristic's maximal finite support, whereas its cumulative rows use the literal 341-occupation common support. An audit found that an earlier version of this module had retained historical full-static-panel exposure groups. The reported module now authenticates and uses the canonical corrected-preperiod membership and Webb normalization throughout; its 468-occupation baseline exactly reproduces -0.1321. The support-specific computer comparison retains 455 occupations and holds those rebuilt primary-universe groups fixed, while the focused exercise requires both computer-use and remotability data and rebuilds groups on its 408-occupation support. Their coefficients therefore differ by design, not by transcription. Every addition and this corrective rerun are labeled post-outcome exploratory.

D.3 Pandemic shortfall

The pandemic-shortfall characteristic is constructed without log pseudocounts. For each occupation, a linear level trend is fit to total employment in 2017–2019; the characteristic is the average 2020–November-2022 level residual relative to that trend. December 2022 remains the transition month and is not used. The measure describes total occupational employment rather than the young-to-older ratio. It is generated from CPS outcomes and may share sampling error and regression-to-the-mean with the dependent stocks. The main multiplier interval holds it fixed; the sampling audit therefore treats this row as diagnostic and does not call it an exogenous control.

D.4 Industry, education, and exact age

The industry analysis is rebuilt at the occupation-by-broad-industry-by-age-by-month level rather than assigning each occupation a dominant industry. The risk set covers all 468 occupations and 3,127 preperiod-connected occupation–industry cells. Relative to the stratified baseline coefficient of -0.1376, adding 13 broad-industry-specific young-by-post shifts gives

TABLE 9. Post-outcome exploratory characteristic-conditioning estimates

Conditioning block	Occupations	Baseline coefficient	Augmented coefficient [95% CI]	Paired change [95% CI]	Paired MDE_{80}
Panel A. One-at-a-time maximal-support comparisons					
Computer use	455	−0.096	−0.197 [−0.316, −0.077]	−0.101 [−0.176, −0.025]	0.108
Remotability	408	−0.115	−0.122 [−0.222, −0.022]	−0.007 [−0.048, 0.033]	0.058
Preperiod wage	457	−0.144	−0.161 [−0.248, −0.075]	−0.017 [−0.044, 0.009]	0.038
Education requirement	404	−0.100	−0.105 [−0.203, −0.008]	−0.005 [−0.027, 0.016]	0.031
Routine-task intensity	449	−0.137	−0.129 [−0.221, −0.036]	0.008 [−0.011, 0.027]	0.027
Manual/physical content	455	−0.140	−0.142 [−0.289, 0.004]	−0.002 [−0.092, 0.089]	0.130
Total-employment pandemic shortfall	468	−0.132	−0.133 [−0.221, −0.044]	−0.0005 [−0.0014, 0.0005]	0.0013
Young-relative pandemic shortfall	458	−0.132	−0.129 [−0.216, −0.041]	0.003 [−0.005, 0.012]	0.012
SOC2 by young by post	468	−0.132	−0.022 [−0.162, 0.119]	0.111 [0.008, 0.213]	0.146
Panel B. Cumulative comparisons on the 341-occupation common support					
Computer use	341	−0.124	−0.248 [−0.391, −0.106]	−0.125 [−0.211, −0.038]	0.124
+ remotability	341	−0.124	−0.250 [−0.388, −0.111]	−0.126 [−0.210, −0.042]	0.121
+ wage, education, routine, and manual	341	−0.124	−0.233 [−0.390, −0.077]	−0.110 [−0.221, 0.002]	0.164
+ total-employment pandemic shortfall	341	−0.124	−0.226 [−0.380, −0.073]	−0.103 [−0.212, 0.007]	0.161
+ SOC2 by young by post	341	−0.124	−0.107 [−0.260, 0.046]	0.017 [−0.116, 0.150]	0.192
Parsimonious block plus SOC2	341	−0.124	−0.124 [−0.268, 0.020]	−0.000004 [−0.125, 0.125]	0.181

Notes: All entries are post-outcome exploratory and were designed after the initial outcome pattern was known. Each baseline–augmented pair uses identical occupations, the canonical rebuilt corrected-preperiod beta groups and Webb normalization, and 9,999 common occupation-level Rademacher multipliers. Panel A uses each characteristic's maximal finite subset; Panel B uses the literal all-characteristic common support, which covers 76.77 percent of primary-support January 2017–November 2022 treatment-construction stock. For comparison, the unaugmented coefficient on all 468 occupations is −0.132, while the unaugmented coefficient after imposing the 341-occupation common support is −0.124; that support-only movement is kept separate from every paired conditioning change. The parsimonious block contains computer use, remotability, education requirement, routine-task intensity, total-employment pandemic shortfall, and SOC2 interactions. Pandemic shortfalls are generated from related CPS outcomes and are held fixed in the reported resampling. A confidence interval containing zero is a nondetection, not evidence of equivalence; conditioning does not identify a causal AI component.

TABLE 10. Post-outcome exploratory support and target-information diagnostics

Conditioning block	Employment coverage (%)	Information retained (%)	Target VIF-like ratio	Effective occupation count	Top-five information share (%)
Panel A. One-at-a-time maximal-support comparisons					
Computer use	96.75	29.02	3.45	52.58	22.34
Remotability	88.58	52.66	1.90	29.25	30.65
Preperiod wage	97.93	59.45	1.68	37.86	26.32
Education requirement	84.54	63.60	1.57	27.48	31.24
Routine-task intensity	98.45	65.19	1.53	43.26	24.59
Manual/physical content	96.85	30.61	3.27	51.05	22.34
Total-employment pandemic shortfall	100.00	69.85	1.43	43.24	24.51
Young-relative pandemic shortfall	99.95	69.71	1.43	43.69	24.20
SOC2 by young by post	100.00	20.81	4.81	53.06	21.87
Panel B. Cumulative comparisons on the 341-occupation common support					
Computer use	76.77	29.33	3.41	48.88	21.98
+ remotability	76.77	28.46	3.51	44.97	24.72
+ wage, education, routine, and manual	76.77	21.08	4.74	58.00	20.28
+ total-employment pandemic shortfall	76.77	20.92	4.78	59.09	19.75
+ SOC2 by young by post	76.77	13.84	7.22	64.36	18.59
Parsimonious block plus SOC2	76.77	16.18	6.18	58.83	20.70

Notes: All diagnostics are post-outcome exploratory. Employment coverage is the share of primary-support January 2017–November 2022 corrected treatment-construction stock retained. "Information retained" is nuisance-adjusted target information divided by fixed-effect-adjusted raw target information; its reciprocal is the reported target VIF-like ratio. These are design-information diagnostics, not an estimate of latent exposure reliability. The effective occupation count is the inverse concentration of occupation-level conditional target-information shares, and the final column is the share supplied by the five largest occupations. In the 341-occupation common-support sample, the 2017–2019-employment-weighted correlations of beta exposure with computer use, remotability, wage, education requirement, routine-task intensity, manual/physical content, total-employment shortfall, and young-relative shortfall are 0.807, 0.562, 0.429, 0.421, 0.197, −0.748, −0.216, and 0.026, respectively. Substantial information loss makes residual coefficients less precise and does not purify an AI treatment.

-0.0987. The paired movement is 0.0389, with interval $[0.0082, 0.0697]$ and paired $\text{MDE}_{80} = 0.0444$. The conditioned coefficient remains below zero. These are statements about the microcell objective, not firm-level adoption.

Education-specific models are estimated on a common occupational support. The BA-or-higher estimate is -0.1380 and the non-BA estimate is -0.1226; their paired difference is -0.0154, with interval $[-0.1557, 0.1250]$ and paired $\text{MDE}_{80} = 0.2014$. The comparison is too imprecise to distinguish moderate education differences. Exact-age estimates for ages 22, 23, 24, and 25 are reported as one jointly estimated exploratory collection with simultaneous intervals. No single age is selected after inspection.

TABLE 11. Industry, education, and exact-age comparisons

Model	Occupations	Coefficient	Pointwise 95% interval	Simultaneous 95% interval
Industry-cell baseline	468	-0.138	$[-0.228, -0.048]$	—
Industry-conditioned	468	-0.099	$[-0.188, -0.009]$	—
BA or higher	429	-0.138	$[-0.275, -0.001]$	$[-0.293, 0.017]$
No BA	429	-0.123	$[-0.231, -0.014]$	$[-0.243, -0.002]$
Age 22	439	-0.093	$[-0.212, 0.026]$	$[-0.239, 0.053]$
Age 23	439	-0.140	$[-0.252, -0.028]$	$[-0.280, -0.0003]$
Age 24	439	-0.222	$[-0.358, -0.086]$	$[-0.394, -0.050]$
Age 25	439	-0.082	$[-0.188, 0.024]$	$[-0.212, 0.047]$

Notes: Industry models use occupation–industry–month cells. Education and age collections use their own common occupational support and 9,999 common occupation multipliers. The four-age common-score equality test gives $\chi^2(3) = 8.21$, $p = .0419$, but every simultaneous age-minus-pooled interval includes zero; no individual age departure is promoted.

E Inference and Finite-Sample Audit

E.1 Stochastic targets

The principal wild-score interval treats occupation as the shock cluster and conditions on the constructed stocks, exposure labels, occupational bridge, and retained support. It accommodates arbitrary serial dependence within an occupation but not unrestricted dependence across occupations. A 22-family Webb-weight exercise instead allows common shocks within broad SOC families. A separate household/sample-unit multiplier rebuilds cell stocks from microdata, preserves all observed records of the sampled unit across months, and assigns the same multiplier to every fractional descendant of a routed source record. That exercise describes repeated-sample sensitivity conditional on released CPS final weights; the public

Basic Monthly extracts do not supply the design variables needed to call it complete CPS design-based inference.

These variances are not mechanically added. Occupation-shock and household-resampling procedures can contain overlapping components, so summing them without a derived orthogonal decomposition could double count uncertainty. They are reported as separate sensitivity targets.

E.2 Score and bootstrap implementation

The conditional likelihood retains one-sided zero cells and drops only occupation–month cells with zero total stock. Scores use the fitted grouped-binomial objective, and nuisance-adjusted target scores use the Schur-complement projection implied by the fitted information matrix. The same 9,999 Rademacher occupation multipliers are reused for paired comparisons. Each draw first forms the two refitted or score-updated targets and then their difference; intervals for movement therefore retain covariance. The finite-cluster correction, random seed, multiplier matrix hash, convergence checks, and score-conservation checks are written to the execution receipts.

For the reconstructed pooled comparison, the analytic standard error is 0.04517 and the 9,999-draw 95-percent interval is $[-0.2206, -0.0437]$. After SOC2-by-calendar-month conditioning, the coefficient is -0.0217, with standard error 0.07132 and interval $[-0.1607, 0.1173]$. The paired movement is 0.11043, with standard error 0.05189 and interval $[0.0107, 0.2102]$. The corresponding two-sided normal-theory paired MDE_{80} is 0.14539. The detected movement remains smaller than this planning quantity; that is not a contradiction because an MDE is a power summary, not an acceptance threshold for an observed draw.

E.3 Broad-family and household sensitivity

The few-family exercise uses 22 broad-family weights and reports baseline, conditioned, and paired targets from common draws. Its role is sensitivity to cross-occupation dependence, not a declaration that SOC2 is the uniquely correct cluster. The household/sample-unit exercise performs 199 positive-weight full refits. Under reconstructed exposure labels, its percentile interval is $[-0.1794, -0.0800]$ for the pooled coefficient, $[-0.1119, 0.0560]$ for the family-conditioned coefficient, and $[0.0370, 0.1722]$ for their movement. Labels and support remain fixed because the target is repeated-sample uncertainty conditional on the preperiod construction; regenerating the generated treatment would define a broader and much more expensive target.

The source does not expose official Basic Monthly replicate weights or all confidential design identifiers. Household identifiers are therefore a dependence-preserving public-data approximation, not an ASEC replication-weight procedure. Link-specific flow analyses additionally report person- and household-cluster sensitivities using their own risk sets.

E.4 Calendar-time HAC and stress validation

The time-HAC audit operates on nuisance-adjusted score sums and implements occupation clustering plus an aggregate elapsed-calendar HAC term minus within-occupation elapsed-calendar HAC terms. Calendar lags are measured in months, so the missing October 2025 observation does not make September and November adjacent. Under rebuilt exposure labels, lag-16 standard errors are 0.04449 for the pooled target, 0.05522 for the family-conditioned target, and 0.04501 for their paired movement. None of the 15 rebuilt five-parameter covariance matrices across the three targets and five lags is indefinite, and no projection or eigenvalue clipping is applied. Indefinite positive-lag matrices from the superseded historical-treatment/SOC2-post audit are retained only as labeled implementation history.

A sparse-cell stress simulation introduces concentrated occupational influence and broad-family shocks under a zero target. In that deliberately adverse design, the nominal occupation-cluster normal interval rejects in 26.7 percent of simulations for the pooled model and 11.3 percent for the family-conditioned model. These rates are not estimates of actual CPS coverage. They show that 9,999 draws control Monte Carlo noise without validating the asymptotic reference distribution, and they motivate reporting the alternative stochastic targets together.

F Dynamics, Endpoints, and Trend Sensitivity

F.1 Quarterly system

The corrected dynamic model interacts every nonreference exposure quintile with calendar quarter. The omitted period is 2022Q4, which contains October and November because December 2022 is the transition month and is excluded. The resulting Q5–Q1 path has 23 preperiod and 15 postperiod coefficients. The pooled and SOC2-by-calendar-month versions are estimated on identical reconstructed support, and the archive stores the complete Q2–Q5 coefficient vector, full covariance matrix, influence matrix, and simultaneous critical values.

The joint null that all 23 Q5 preperiod coefficients equal zero is rejected in both the pooled model ($p = .0149$) and the family-conditioned model ($p = .0053$). As a precision diagnostic, an OLS linear functional through those coefficients is -0.0078 log point per year in the pooled model and 0.0041 after family conditioning. Its occupation-cluster standard errors are 0.0119 and 0.0292, implying normal-theory MDE_{80} values of 0.0332 and 0.0819 log point per year. Thus an insignificant linear slope cannot rule out economically relevant smooth drift and is not a substitute for the rejected unrestricted joint test. Simultaneous bands exclude zero only in late pooled quarters and never in the family-conditioned path. These results do not establish which preperiod force produces the differences; they make a parallel-evolution interpretation of the January 2023 split untenable.

The postperiod dynamic functional averages the 15 post-quarter coefficients with predeclared calendar weights. It equals -0.1199 in the pooled model, with interval $[-0.2618, 0.0221]$, and -0.2074 in the family-conditioned model, with interval $[-0.4216, 0.0067]$. This linear functional is not the nonlinear static grouped-binomial coefficient. The mapping file reports their paired relationship rather than silently treating them as identical.

F.2 Official HonestDiD implementation

The trend-sensitivity program passes the joint event-time vector and covariance matrix to the official `HonestDiD` implementation. The environment receipt fixes HonestDiD 0.2.8 and its source commit, and records the compatible official CVXR and solver versions. The program does not feed a single static coefficient and standard error into an event-study procedure. The conventional event-functional interval already includes zero for both the pooled and family-conditioned vectors. The correct zero-exclusion breakdown is therefore zero: there is no positive robustness threshold to lose. Earlier installation and API failures remain in the execution log; the successful official run is not replaced with a home-built approximation.

F.3 Onset, endpoint, seasonality, and survey dates

Onset sensitivity reports every monthly split from November 2022 through June 2023. January 2023 is retained because December 2022 was designated as the transition month after the November 30, 2022 public release of ChatGPT; robustness across nearby dates does not identify that event as the cause. The endpoint grid includes December 2024, September and December 2025, and July 2026 wherever the calendar permits. October 2025 is never imputed. Extending from the 2025 endpoints to July 2026 makes the pooled coefficient about 0.02 log point more negative, and its paired interval excludes zero; the analogous conditioned movements remain near zero.

Exposure-group-by-month-of-year interactions supply the lower-dimensional seasonality check and move the target little. More saturated occupation-specific seasonality and a shortened

TABLE 12. Official HonestDiD sensitivity intervals

Structure	Restriction	Lower	Upper
Pooled	Conventional event-functional interval	−0.264	0.024
Pooled	Smoothness, $M = 0$	−0.140	0.016
Pooled	Smoothness, $M = 0.005$ per quarter	−0.482	0.516
Pooled	Relative magnitude, $\bar{M} = 0$	−0.260	0.019
Pooled	Relative magnitude, $\bar{M} = 0.5$	−1.023	0.789
SOC2 monthly paths	Conventional event-functional interval	−0.425	0.010
SOC2 monthly paths	Smoothness, $M = 0$	−0.351	0.007
SOC2 monthly paths	Smoothness, $M = 0.005$ per quarter	−0.869	0.352
SOC2 monthly paths	Relative magnitude, $\bar{M} = 0$	−0.420	0.007
SOC2 monthly paths	Relative magnitude, $\bar{M} = 0.5$	−2.222	1.884

Notes: Intervals use the reconstructed quarterly Q5–Q1 vector, its full occupation-cluster covariance, and the observed-calendar post-quarter functional. Smoothness M is in log points per quarter; $\bar{M}$ bounds post violations relative to the largest preperiod violation. The conventional interval already includes zero in both structures, so positive zero-exclusion robustness is undefined (reported as zero), not an omitted threshold.

TABLE 13. Post-outcome exploratory endpoint sensitivity

Endpoint	Months	Coefficient [95% CI]	Endpoint − full [paired 95% CI]	Paired MDE_{80}
Pooled				
Through December 2024	95	−0.1110 [−0.2008, −0.0213]	0.0211 [−0.0073, 0.0495]	0.0407
Through September 2025	104	−0.1096 [−0.1978, −0.0214]	0.0225 [0.0053, 0.0397]	0.0245
Through December 2025 (actual gap)	106	−0.1109 [−0.1979, −0.0239]	0.0212 [0.0057, 0.0367]	0.0221
Through July 2026, excluding September and November 2025	111	−0.1337 [−0.2236, −0.0438]	−0.0016 [−0.0075, 0.0043]	0.0084
Through July 2026 (reference)	113	−0.1321 [−0.2206, −0.0437]	Reference	—
SOC2 by calendar month				
Through December 2024	95	−0.0120 [−0.1657, 0.1417]	0.0097 [−0.0526, 0.0719]	0.0897
Through September 2025	104	−0.0219 [−0.1616, 0.1178]	−0.0002 [−0.0345, 0.0341]	0.0491
Through December 2025 (actual gap)	106	−0.0170 [−0.1555, 0.1215]	0.0047 [−0.0248, 0.0342]	0.0422
Through July 2026, excluding September and November 2025	111	−0.0252 [−0.1660, 0.1155]	−0.0035 [−0.0131, 0.0060]	0.0137
Through July 2026 (reference)	113	−0.0217 [−0.1607, 0.1173]	Reference	—

Notes: All rows use the rebuilt corrected-preperiod treatment and the same occupation support. The December 2024 row is the requested pre-2025 endpoint; the December 2025 row retains the actual missing October survey month. The shutdown-month row excludes September and November 2025 and does not interpolate October. Nonreference endpoint intervals and paired differences use the common endpoint-grid draws. To avoid printing a second seed-dependent interval for an identical target, each full-window reference uses its declared canonical single-target interval. The MDE is the two-sided five-percent, 80-percent normal-theory precision diagnostic, not an economic-equivalence threshold.

post-2020 fit fail to converge under the declared algorithm; both attempts and diagnostics are retained. Excluding September and November 2025 around the collection gap does not detectably change the full-window estimate. These rows address sensitivity to calendar construction and survey production, not counterfactual reweighting for the January 2025 or January 2026 population controls.

F.4 Official-weight and respondent-equivalent era comparison

Table 14 reports a post-outcome exploratory joint-model comparison of 2023–2024 with 2025–2026. The official-weight row retains the paper's employment-stock estimand; the respondent-equivalent row is a deliberately different routed-count outcome used only to diagnose whether the later pattern mechanically appears only in final-weight stocks. It cannot recover unpublished age-by-occupation counterfactual weights.

TABLE 14. Post-outcome exploratory comparison of early and later postperiods

Cell construction	2023–2024 coefficient [95% CI]	2025–2026 coefficient [95% CI]	Later minus early [95% CI]	Paired MDE_{80}
Official final-weight stocks	−0.111 [−0.197, −0.024]	−0.166 [−0.270, −0.062]	−0.056 [−0.123, 0.012]	0.097
Respondent-equivalent routed counts	−0.103 [−0.193, −0.012]	−0.183 [−0.277, −0.089]	−0.080 [−0.139, −0.021]	0.085

Notes: This timing comparison was produced after outcomes were opened and is post-outcome exploratory. Both rows use the corrected 113-month calendar, the same 468 occupations, a joint two-era specification, and 9,999 common occupation multipliers within each paired later-minus-early contrast. Official final-weight stocks are the paper's employment-stock object. Respondent-equivalent values are fractional routed-record counts after occupational crosswalking; they are neither distinct survey respondents nor a recovered counterfactual CPS weight vintage. The official-weight interval does not detect an era difference, whereas the respondent-equivalent interval excludes zero. That divergence does not identify response composition or show that population-control revisions caused the later estimate.

G Linked Flows and Worker Outcomes

G.1 Link construction and risk sets

Person links follow official IPUMS/Census matching guidance and use the available longitudinal weight for each eligible link. Adjacent-month and twelve-month endpoints are distinct samples. A twelve-month link is not the sum of twelve observed monthly transitions and may conceal intervening changes. The annual young link rate is 50.98 percent, compared with 70.55 percent for the older group, so the longer horizon entails appreciable differential selection.

Employment exit begins with an employed person in an occupation on the analysis support. Reported occupation change requires employment at both endpoints and a valid occupation at each. Entry-destination allocation begins with a nonemployed origin, ends with observed employment, and asks whether the destination falls in Q5 rather than Q1. It therefore conditions on entry and is neither an employment-finding probability nor an employer hiring rate. Every crosswalk descendant of one source record shares its link and resampling weight.

G.2 Results and uncertainty

The adjacent-month young-relative Q5–Q1 estimates are 0.132 for employment exit, 0.003 for reported occupation change, and -0.070 for entry-destination allocation. Their 95-percent intervals are respectively $[-0.042, 0.307]$, $[-0.118, 0.123]$, and $[-0.250, 0.110]$. Twelve-month endpoint estimates are 0.123, -0.018, and -0.052, with intervals $[-0.111, 0.357]$, $[-0.097, 0.061]$, and $[-0.251, 0.147]$. All intervals include zero. Person- and household-cluster sensitivity results, descriptive transition rates, effective risk counts, and named occupation influence appear in the machine-readable flow directory.

Usual-hours changes are estimated within linked employed persons. The adjacent estimate is -0.119 hours per week with interval $[-0.295, 0.057]$. Weekly earnings use the eligible outgoing-rotation sample and its earnings weight; the conditional log-earnings coefficient is -0.0245 with interval $[-0.1024, 0.0534]$. The earnings extract covers 74 months through March 2023 and therefore contains only three postperiod months. Both outcomes condition on employment. Unemployment duration is described only among persons observed unemployed, and no occupation is imputed to never-employed or out-of-labor-force respondents. These restrictions prevent the outcomes from being misreported as population employment effects.

TABLE 15. Employment-exit components in linked CPS records

Endpoint	Component	Raw events	Coefficient	95% interval
Adjacent month	Unemployment entry	32,488	0.055	$[-0.272, 0.381]$
Adjacent month	Labor-force exit	74,480	0.135	$[-0.064, 0.333]$
Twelve month	Unemployment at endpoint	27,330	0.246	$[-0.295, 0.787]$
Twelve month	Out of labor force at endpoint	78,773	0.078	$[-0.181, 0.337]$

Notes: Rows use official longitudinal weights and the same young-relative Q5–Q1 log-rate-ratio contrast as the core flow table. Counts are repeated link events rather than unique persons. Unemployment and out-of-labor-force events exhaust observed employment exits, but their nonlinear coefficients do not add. Every interval includes zero.

G.3 Why no stock–flow calibration is reported

A compatible accounting identity would require entries from nonemployment, exits, switches among all occupations, aging into and out of ages 22–25, corresponding movements for the older denominator, job-to-job moves, and population/sample-composition residuals. The CPS endpoint links do not measure all of these components on a common employer-match risk set, and the BCC hiring and separation objects are not identical to CPS employment transitions. A single implied stock response would therefore be assumption driven. The revision reports the measurable margins and the precise comparability gap instead of manufacturing a calibration.

H Payroll Bridge and Exposure-Construction Diagnostics

H.1 BCC version and grouping bridge

The comparison uses the August 12, 2026 version of Brynjolfsson–Chandar–Chen and records the accessed paper, dashboard, and workbook in the evidence ledger. Their reported 19-percent young-worker "kept pace" shortfall is a stock-growth counterfactual in ADP data, not a regression coefficient, employment probability, or hiring rate. Their public dashboard forms equal-occupation exposure groups; later regressions employ other weights and employer controls. Exact dashboard memberships and the treatment of ties are not released.

The CPS bridge therefore applies an equal-occupation, tie-preserving top-two-versus-bottom-three approximation on the present support and labels it as such. The pooled estimate is -0.0720, with interval $[-0.1218, -0.0222]$. The SOC2-by-calendar-month estimate is -0.0168, with interval $[-0.0810, 0.0474]$. Their paired movement is 0.0553, with interval $[-0.0024, 0.1129]$. The archive provides group summaries and a membership file; it cannot provide concordance with undisclosed assignments. Differences in occupation universe, age comparison, denominator, endpoint, firm controls, and employer-matched flows remain.

H.2 Task-capability weighting

Write the two Eloundou primitives as $D = E1$ and $S = E2$. The score family $D + \lambda S$ is evaluated on a fixed grid from zero to one under common preperiod weights and common bootstrap draws. The implementation verifies numerically that $\lambda = 0.5$ reproduces raw beta before grouping. Any earlier mismatch between that row and the headline arose from a different support or treatment contract and is preserved only as implementation history.

Two versions are kept separate. The native-rank analysis recomputes the score normalization and tie-preserving groups at each λ, so both scale and membership can change. The fixed-membership analysis changes only the continuous score representation while holding

TABLE 16. Post-outcome exploratory BCC-style grouping bridge

Grouping rule	Conditioning structure	Coefficient [95% CI]	Change from baseline [95% CI]	Paired MDE_{80}
Panel A. Public-dashboard equal-occupation approximation				
Equal occupation	Occupation and calendar-month FE	-0.072 $[-0.122, -0.022]$	—	—
Equal occupation	SOC2 by young by post	-0.015 $[-0.080, 0.050]$	0.057 $[-0.001, 0.115]$	0.083
Equal occupation	SOC2 by young by calendar month	-0.017 $[-0.081, 0.047]$	0.055 $[-0.002, 0.113]$	0.082
Panel B. Historical employment-weighted approximation				
Employment weighted	Occupation and calendar-month FE	-0.075 $[-0.125, -0.025]$	—	—
Employment weighted	SOC2 by young by post	-0.027 $[-0.095, 0.041]$	0.048 $[-0.011, 0.108]$	0.086
Employment weighted	SOC2 by young by calendar month	-0.027 $[-0.093, 0.040]$	0.048 $[-0.011, 0.107]$	0.086

Notes: All rows are post-outcome exploratory CPS bridges, not replications of BCC's proprietary ADP analysis. Each row compares GPT-4-beta Q4–Q5 with Q1–Q3 on the same 468 occupations, includes the standardized historical Webb-software control, and uses 9,999 common occupation-level Rademacher multipliers. Paired changes compare each SOC2-conditioned row with its own unconditioned baseline. Panel A follows the public dashboard's equal-occupation description as closely as public information permits. Panel B uses earlier employment-weighted cuts formed from 108 months of young-plus-older CPS stock, including postperiod months, and is only a grouping-rule diagnostic. Exact BCC occupation membership, universe, cutoff and tie algorithm are not public, so concordance is unverified. The BCC and CPS objects also differ in ages, denominator, endpoint, employer controls, and observed worker–firm flows. Null-containing paired intervals are nondetections, not equivalence.

the beta groups fixed. For every grid point the result files report coefficient, paired difference from beta, interval, occupation coverage, tail turnover, and correlations with computer use and remotability. A focused 408-occupation run adds both computer use and remotability on common support. None of these paths measures realized deployment.

H.3 Primitive and architecture comparisons

The joint primitive model reports D and S in raw share units, in preperiod-standard-deviation units, their covariance, and an illustrative common change. This is a two-regressor parameterization, not a causal decomposition into technologies. Architecture comparisons likewise report both component coefficients and their paired difference on each declared support. Webb-AI comparisons are shown first on identical support and then on broader native support so that score changes are not confused with population changes. Principal-component rotations and earlier F/G coordinates remain in the replication archive because they are affine re-expressions; they are omitted from the scientific appendix.

TABLE 17. Post-outcome exploratory direct and software-complement primitives

Units or contrast	Term	Estimate [95% CI]	Bootstrap SE	MDE_{80}
Raw share	$D\times$ young $\times$ post	-0.189 [-0.354, -0.024]	0.085	0.239
Raw share	$S\times$ young $\times$ post	-0.089 [-0.172, -0.005]	0.043	0.121
Preperiod SD	$D\times$ young $\times$ post	-0.031 [-0.058, -0.004]	0.014	0.039
Preperiod SD	$S\times$ young $\times$ post	-0.028 [-0.054, -0.002]	0.014	0.038
One SD in each primitive	$D+S$ joint change	-0.059 [-0.098, -0.020]	0.020	0.056

Notes: All entries are post-outcome exploratory and use the same 468 occupations and 9,999 common occupation multipliers. The joint model includes D, S, and standardized Webb software, each interacted with young $\times$ post. Here $D = E1$ is the share of tasks classified as directly accelerable and $S = E2$ is the share requiring complementary software; beta satisfies $D + 0.5S$. The preperiod employment-weighted means (standard deviations) are 0.144 (0.164) for D and 0.463 (0.315) for S. The common-draw correlation between the two standardized coefficient draws is 0.062. Component-row MDEs use the analytic occupation-clustered standard errors; the joint-change MDE uses its common-draw standard error. The final row is the declared one-standard-deviation increase in both primitives, not a paired difference between two technologies. The parameterization does not measure adoption and is not a causal decomposition.

H.4 What validation would require

None of the sampling intervals above treats a score as a noisy observation of a validated true exposure. A useful validation sample would begin at the versioned occupation–task level and oversample both broad families and tasks on which architectures disagree. It would record

TABLE 18. Post-outcome exploratory task-weight architectures on common support

Architecture pair	Alternative coefficient	Beta coefficient	Alternative − beta [95% CI]	Paired MDE_{80}
D versus $D + 0.5S$	−0.103	−0.132	$0.029\ [-0.044, 0.103]$	0.106
$D + 0.25S$ versus $D + 0.5S$	−0.101	−0.132	$0.031\ [-0.005, 0.068]$	0.055
$D + 0.75S$ versus $D + 0.5S$	−0.143	−0.132	$-0.011\ [-0.039, 0.017]$	0.041
$D + S$ versus $D + 0.5S$	−0.156	−0.132	$-0.024\ [-0.064, 0.017]$	0.058

Notes: All entries are post-outcome exploratory Q5–Q1 coefficients estimated on the identical 468-occupation support. Beta is $D + 0.5S$. At each alternative weight, normalization, cutoffs, and tie-preserving quintile memberships are recomputed, so a paired movement combines score and membership changes rather than holding the treatment groups fixed. The intervals use 9,999 common occupation multiplier draws, preserving covariance within each pair. Every paired interval contains zero; this means that the design does not detect a difference, not that the architectures are economically equivalent.

TABLE 19. Post-outcome exploratory Webb conditioning and support

Specification	Occupations	Beta coefficient [95% CI]	Same-support paired change [95% CI]
Without Webb, fixed primary support	468	$-0.134\ [-0.219, -0.049]$	$-0.002\ [-0.010, 0.007]$
With Webb, fixed primary support	468	$-0.132\ [-0.220, -0.044]$	Reference
Without Webb, broader beta-valid support	490	$-0.135\ [-0.218, -0.051]$	Not paired

Notes: All entries are post-outcome exploratory. The first two rows hold occupations, beta cutoffs, and the outcome specification fixed; the displayed paired change is without-Webb minus with-Webb, uses common draws, and has $\mathrm{MDE}_{80} = 0.013$. Its interval containing zero is a nondetection, not equivalence. The final row relaxes Webb availability and therefore changes support and beta cutoffs; it has no paired interpretation. Webb conditioning is an occupational-software proxy, not realized software or AI adoption.

separate outcomes for technical capability, realized time savings without complementary software, realized time savings with software, observed workplace adoption, and task importance. Conflating these labels would reproduce the construct problem the validation is meant to solve.

Labels should come from repeated, blinded, independent expert assessments and, where feasible, audited product-use or task-performance records. Calibration and evaluation occupations should be separated so that crosswalk and occupation aggregation choices are tested out of sample. The task description, model snapshot, prompt, complementary tools, rater instructions, occupational bridge, and employment weights all require version records. Sample size cannot be stated as one universal number: it depends on within-occupation task correlation, rater reliability, unequal task weights, the latent quantity being corrected, and whether the downstream target is a continuous slope or tail assignment. In the absence of these inputs, the revision specifies the validation design but does not apply a generated-regressor correction.

I Reproducibility, Numerical Consistency, and Unresolved Items

I.1 Execution contract

The revision archive separates restricted inputs from distributable code and derived aggregate outputs. Two commands serve different purposes. The restricted-data orchestrator authenticates the licensed inputs, refuses to overwrite an existing output root, runs the rebuilt baseline before dependent modules, passes named treatment objects downstream, invokes each self-check, and runs the pinned official HonestDiD implementation. A separate aggregate-only command regenerates exhibits from committed results, checks numerical and prose consistency, runs repository tests, and builds the LaTeX documents. The latter neither reads CPS microdata nor independently re-estimates the empirical models. Environment records report Python and R package versions, seeds, scheduler resources, source hashes, and git state where captured. Every module declares its treatment contract, calendar, support, objective, inference target, and output paths before interpreting the result.

The Git repository is not itself labeled a sanitized public-release artifact. A machine-readable positive allowlist defines the files eligible for staging; unlisted files and suffixes are excluded. The release scan rejects private absolute roots, credential-token patterns, restricted-data formats, symlinks, caches, and tabular columns containing direct CPS link identifiers, followed by manual disclosure and metadata review. Licensed CPS records, the historical private access receipt, account metadata, download links, longitudinal identifier values, credentials, and operational path history remain outside the staged package. The manifest records IPUMS extract numbers 9, 10, and 11 and the available request, data, DDI, and codebook hashes; unavailable API timestamps are left blank rather than inferred.

The canonical results ledger is the numerical source of truth. Manuscript tables are generated from or checked against it, and repeated estimates use one declared draw set. If an interval comes from a different stochastic target—occupation shocks, broad-family shocks, or household resampling—the table labels that procedure rather than presenting multiple intervals as interchangeable versions of one result. The test suite verifies calendar membership, one-sided zeros, route conservation, fixed-support pairing, covariance symmetry and rank, draw hashes, and table/prose consistency.

I.2 Implemented failures

The failure registry is part of the evidence. It includes the literal vintage merge, the original March sample selection, nonconvergence of saturated seasonality variants, indefinite positive-lag joint HAC matrices from the superseded historical-treatment audit, and unsuccessful software installation attempts. A failure is never converted into a different estimator without an explicit amendment. Successful reruns preserve the earlier log and record the replacement's package and solver versions.

I.3 Unresolved limitations

The analysis cannot observe firm adoption, employer identifiers, worker–firm hiring, confidential CPS design variables, exact BCC dashboard memberships, age-specific occupational route probabilities, or a validated latent "true" AI exposure. The public data therefore do not identify an AI causal effect or a complete stock–flow mechanism. The direct within-family tail population is small, unrestricted preperiod differences are present, several linked-flow intervals are wide, and January 2025–2026 survey-production changes cannot be undone with published aggregate factors. These are limits on interpretation, not missing robustness rows that a specification search can solve.

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